Meric Altug Gemalmaz, Ph.D.

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EDUCATION

Purdue University, West Lafayette, Indiana

- Ph.D. in Computer Science (Human-AI Interaction Focused) **3.96/4.0** — Jan. 2020 – June 2025
- M.S. in Computer Science (Transferred to Ph.D.) **3.93/4.0** — Jan. 2019 – Dec. 2019
- B.S. in Computer Science (Software Engineering Track, **Distinction**) **3.92/4.0** — Aug. 2015 – Dec. 2018

SKILLS

Programming Languages: Python, C/C++, C#, JavaScript, Java, HTML/CSS, R, SQL, Bash
Machine Learning: Large-scale retrieval and ranking, recommender systems, content-integrity and quality signals, LLM-based signal design and prompting, model evaluation, A/B experimentation and metric design
Machine Learning Tools: scikit-learn, GSL, NumPy, Pandas
Tools & Platforms: Git, Meteor.js, MongoDB, MTurk, Android SDK, Firebase, GDB

EXPERIENCE

ML Research Scientist, Meta, Menlo Park, CA — Aug. 2025 – Present

- Contributed to large-scale recommendation and ranking systems, first on content recommendation and content quality for Facebook (FB) Groups, now on data and evaluation for AI agent systems.
- Shipped **17** A/B-tested launches in roughly nine months on FB Groups, across **two** distinct recommendation surfaces and tech stacks: built LLM-based content-quality and content-integrity signals, plus **two** new candidate generators.
- Upheld engineering excellence by ramping onto two tech stacks **in my first month of joining**, taking production oncall by **month four**, and always chasing root causes in retrieval and measurement instead of patching symptoms.
- Build the workflows that create and improve training data for AI agent systems, and evaluate those systems' decisions.

Web App Development for Data Collection, Purdue — Jan. 2020 – May 2025

- Coordinated **1,500+** participants across **4+ MTurk** research projects for data collection and analysis.
- Engineered and deployed web apps with **JavaScript** and **Meteor.js**, **MongoDB** back ends, and **HTML**/**CSS** front ends.
- Delivered **90%** participant satisfaction and blocked **80%** of bot attacks via scenario-based, interactive UIs.

Fairness in AI-Driven Gig Work: User-Centric Insights, Purdue — Dec. 2023 – May 2025

- Initiated a study with gig workers on the long-term impact of AI-driven assignments at varying fairness levels, yielding insights for fairer algorithmic job management.

Design Implications of Fairness in Human-AI Interactions, Purdue [P1, P3] — Mar. 2021 – Dec. 2023

- Conducted **six** human-subject experiments on how loan applicants' repeated interactions with an AI loan-approval system affect fairness perceptions and willingness to keep engaging.
- Simulated **1,000** loan applicant interactions with a **Markov Decision Process** in **Python** to estimate experiment parameters; applied **regression analysis** in **R** to real-world data to uncover human-AI interaction trends.
- Found a critical fairness issue: over **75%** of users continue using biased models out of necessity, not perceived fairness, urging a rethink of usage metrics as fairness indicators.

Data Bias Mitigation Algorithm, Purdue [P2] — Mar. 2020 – Mar. 2021

- Used unsupervised learning to detect and mitigate cognitive bias in crowdsourced annotations.
- Applied **probabilistic graphical models** and **Expectation-Maximization** with **GSL** in **C** to reduce annotator bias, improving label accuracy across **100** simulated and **12** real-world tasks.
- Achieved over **10%** higher inferred label accuracy than existing baselines by reducing annotation bias.

HONORS AND AWARDS

Recipient of the Graduate Teaching Award, Purdue — Nov. 2022
Recipient of the Raymond Boyce Graduate Teaching Award, Purdue — Apr. 2020
Dean's List and Semester Honors (**8 Semesters**), Purdue — Aug. 2015 – Dec. 2018

SELECTED PAPERS

[P1] Meric Altug Gemalmaz, Ming Yin. "Understanding Decision Subjects' Fairness Perceptions and Retention in Repeated Interactions with AI-Based Decision Systems." *Proceedings of the 5th AAAI/ACM Conference on AI, Ethics, and Society (AIES)*, Oxford, UK, Aug. 2022.

[P2] Meric Altug Gemalmaz, Ming Yin. "Accounting for Confirmation Bias in Crowdsourced Label Aggregation." *Proceedings of the 30th International Joint Conference on Artificial Intelligence (IJCAI)*, Online, Aug. 2021.

[P3] Meric Altug Gemalmaz, Ming Yin. "Understanding Decision Subjects' Engagement with and Perceived Fairness of AI Models When Opportunities of Qualification Improvement Exist." *arXiv:2410.03126*